Master table for the closing strategy: the forecasts fitted on the 16:00 bar, one scorer, the same days

What this version contains

205 forecasts, every one fitted on the 16:00 bar alone: the per-bar linear arms (ridge, lasso, elastic net on the HAR + calendar and the all-features inputs; the VIX-only, implied-volatility, HAR-ladder and chain-period bucket studies), every per-bar tree rung (the shipped configuration refit every 10 sessions and every session; the random search at both cadences; the Optuna-tuned paths by tuning cadence and trial budget; MSE- and QLIKE-selected twins). Left out (33 rows of the full table): the paper's pooled 48-bar forecasts, their pooled twins and the rest-of-day check rows (exact copies of their per-bar twins). Always short is kept as the benchmark row. Every number is the full table's (experiments/master_table_close.py; every forecast is the plain back-transform f^2B, no recalibration; 866 trade days; point estimates, with the Diebold–Mariano statistic and the HAC t where shown); the columns against the reference still compare with the paper's block-diagonal ridge, which is not itself listed here. The per-bar design has no session-edge duplicates, every fit keeps only the columns identifiable on its own training window, and every tree chunk was computed on one processor class.

Table 1: Forecast accuracy at the 16:00 bar (15:30–16:00). Every forecast is the plain back-transform $f^2 B$ (f the forecast on the adjusted scale $\sqrt{RV/B}$, B the diurnal baseline). QLIKE: against the actual (unwinsorized) realized variance; QLIKE, trimmed: the same forecast against the winsorized training target (diagnostic). % vs R: the mean daily loss difference against the reference R; DM: Diebold–Mariano statistic on that difference, Newey–West variance (Bartlett, lag 6) with the Harvey–Leybourne–Newbold correction; $\overline{RV}/\overline{F}$: mean realized variance over mean forecast; extreme: plain forecasts below PLAIN_FLOOR on every 16:00 session of the arm span. Point estimates. R = reference, H = recommended headline, = identical to another row.

Forecast	QLIKE	QLIKE, trimmed	% vs R	DM	$\overline{RV}/\overline{F}$	extreme (all sess.)
Table A: the same 866 trade days, 2020-01-03 to 2024-04-30						
per-bar linear						
per-bar ridge [`lean_vol_vix`]	0.0985	0.0966	−9.8	−1.94	0.901	7
per-bar elastic net [`all_features`]	0.1006	0.0986	−7.9	−2.10	0.925	4
per-bar lasso [`all_features`]	0.0996	0.0973	−8.8	−2.80	0.914	4
per-bar elastic net [`lean_vol_vix`], pre-tuned	0.0955	0.0937	−12.5	−3.30	0.988	3
per-bar elastic net [`all_features`], pre-tuned	0.0958	0.0942	−12.2	−2.97	0.950	4
per-bar ridge [`lean_vol_vix`], pre-tuned	0.0935	0.0916	−14.4	−3.78	0.976	3
per-bar lasso [`all_features`], pre-tuned	0.0931	0.0914	−14.7	−3.72	0.945	3
per-bar lasso [`lean_vol_vix`]	0.0958	0.0938	−12.3	−3.64	0.947	4
per-bar ridge [`all_features`]	0.1184	0.1164	+8.4	+0.72	0.907	6
per-bar elastic net [`lean_vol_vix`]	0.0968	0.0945	−11.3	−3.58	0.951	3
per-bar lasso [`lean_vol_vix`], pre-tuned	0.0964	0.0947	−11.8	−2.95	0.903	5
per-bar ridge [`all_features`], pre-tuned	0.0975	0.0958	−10.7	−2.35	0.966	4
per-bar elastic net [`baseline`], pre-tuned	0.0967	0.0946	−11.5	−2.41	1.026	0
per-bar lasso [`baseline`], pre-tuned	0.0967	0.0946	−11.5	−2.41	1.026	0
per-bar ridge [`baseline`], pre-tuned	0.0967	0.0946	−11.5	−2.41	1.026	0
per-bar OLS [`baseline`]	0.0967	0.0946	−11.5	−2.41	1.026	0
per-bar ridge [`baseline`]	0.0968	0.0946	−11.3	−2.43	1.041	0
per-bar lasso [`baseline`]	0.0956	0.0934	−12.4	−2.56	1.052	0
per-bar elastic net [`baseline`]	0.0957	0.0936	−12.4	−2.55	1.047	0
VIX-only family						
per-bar ridge [`live_vix_only`]	0.1201	0.1179	+10.0	+0.55	0.944	5
per-bar ridge [`free_vix_only`]	0.1216	0.1194	+11.4	+0.57	0.941	6
per-bar ridge [`free_feasible`]	0.1431	0.1407	+31.0	+0.90	0.950	0
per-bar ridge [`live_vix_rvol`]	0.1156	0.1134	+5.9	+0.40	0.944	6
per-bar ridge [`free_feasible_vol`]	0.1220	0.1198	+11.7	+0.60	0.956	6
per-bar lasso [`free_vix_only`]	0.0959	0.0940	−12.2	−3.48	0.944	4
per-bar lasso [`free_feasible_vol`]	0.0949	0.0930	−13.1	−3.85	0.949	4
per-bar elastic net [`free_vix_only`]	0.0972	0.0950	−10.9	−3.45	0.951	3
per-bar elastic net [`live_vix_rvol`]	0.0965	0.0942	−11.6	−3.65	0.953	3
per-bar elastic net [`free_feasible_vol`]	0.0973	0.0950	−10.9	−3.60	0.946	3
per-bar lasso [`live_vix_only`]	0.0956	0.0937	−12.4	−3.55	0.947	4
per-bar lasso [`live_vix_rvol`]	0.0988	0.0961	−9.5	−3.18	0.948	4
per-bar elastic net [`live_vix_only`]	0.0975	0.0952	−10.7	−3.33	0.951	3
per-bar lasso [`free_feasible`]	0.0965	0.0945	−11.6	−3.44	0.949	0
per-bar elastic net [`free_feasible`]	0.0983	0.0959	−10.0	−3.22	0.953	0
per-bar lasso [`vix_only`]	0.0957	0.0936	−12.3	−2.66	1.054	1
per-bar ridge [`vix_only`]	0.0962	0.0939	−11.9	−2.68	1.012	3
per-bar ridge [`vix_rvol`]	0.0954	0.0930	−12.7	−2.88	0.997	2
per-bar lasso [`vix_rvol`]	0.0944	0.0922	−13.6	−2.92	1.042	1
per-bar elastic net [`vix_only`]	0.0956	0.0932	−12.5	−2.80	1.026	3
per-bar elastic net [`vix_rvol`]	0.0949	0.0925	−13.1	−2.95	1.020	2
per-bar tree (untuned)						
per-bar XGBoost [`lean_vol_vix`]	0.0943	0.0923	−13.7	−3.47	0.800	1
per-bar random forest (dev-tuned) [`all_features`]	0.0998	0.0974	−8.6	−2.27	0.855	3
per-bar LightGBM [`lean_vol_vix`]	0.0946	0.0928	−13.3	−3.15	0.825	2
per-bar random forest [`lean_vol_vix`]	0.0957	0.0935	−12.3	−3.32	0.866	1
per-bar XGBoost [`all_features`]	0.0990	0.0970	−9.3	−2.25	0.730	1
per-bar LightGBM [`all_features`]	0.0983	0.0962	−10.0	−2.36	0.825	2
per-bar random forest (dev-tuned) [`lean_vol_vix`]	0.0948	0.0924	−13.2	−3.31	0.897	2
per-bar random forest [`all_features`]	0.1006	0.0982	−7.9	−2.23	0.784	1
per-bar XGBoost [`baseline`]	0.0999	0.0974	−8.5	−1.61	1.020	0
per-bar random forest [`baseline`]	0.1006	0.0982	−7.9	−1.52	1.025	1
per-bar random forest (dev-tuned) [`baseline`]	0.1012	0.0988	−7.3	−1.32	1.036	2
per-bar LightGBM [`baseline`]	0.1018	0.0993	−6.8	−1.32	1.038	0
per-bar tree (untuned, daily refit)						
per-bar XGBoost, daily refit [`lean_vol_vix`]	0.0932	0.0911	−14.7	−3.68	0.836	1

(continued)

Forecast	QLIKE	QLIKE, trimmed	% vs R	DM	$\overline{RV}/\overline{F}$	extreme (all sess.)
per-bar random forest, daily refit [lean_vol_vix]	0.0938	0.0916	−14.1	−3.78	0.879	2
per-bar random forest (dev-tuned), daily refit [lean_vol_vix]	0.0947	0.0923	−13.3	−3.31	0.904	2
per-bar LightGBM, daily refit [all_features]	0.0961	0.0939	−12.0	−3.11	0.851	2
per-bar LightGBM, daily refit [lean_vol_vix]	0.0939	0.0918	−14.0	−3.49	0.869	1
per-bar XGBoost, daily refit [all_features]	0.0957	0.0935	−12.3	−3.06	0.827	1
per-bar random forest (dev-tuned), daily refit [baseline]	0.1004	0.0979	−8.1	−1.46	1.016	2
per-bar XGBoost, daily refit [baseline]	0.1001	0.0976	−8.4	−1.58	1.021	0
per-bar random forest (dev-tuned), daily refit [all_features]	0.0980	0.0954	−10.3	−2.68	0.880	3
per-bar random forest, daily refit [all_features]	0.0991	0.0966	−9.2	−2.59	0.827	1
per-bar LightGBM, daily refit [baseline]	0.1010	0.0986	−7.5	−1.44	1.030	0
per-bar random forest, daily refit [baseline]	0.0992	0.0968	−9.2	−1.78	1.016	1
per-bar tree (tuned)						
tuned per-bar random forest [lean_vol_vix], MSE-selected	0.0993	0.0971	−9.1	−2.45	0.873	1
tuned per-bar LightGBM [all_features], MSE-selected	0.1005	0.0982	−7.9	−1.73	0.876	0
tuned per-bar LightGBM [lean_vol_vix], MSE-selected	0.1044	0.1021	−4.4	−0.99	0.908	0
tuned per-bar LightGBM [lean_vol_vix], QLIKE-selected	0.0996	0.0977	−8.8	−2.05	0.907	0
tuned per-bar XGBoost [all_features], MSE-selected	0.1030	0.1007	−5.7	−1.25	0.838	0
tuned per-bar XGBoost [all_features], QLIKE-selected	0.1029	0.1006	−5.7	−1.27	0.837	0
tuned per-bar LightGBM [all_features], QLIKE-selected	0.0998	0.0975	−8.6	−1.95	0.847	0
tuned per-bar random forest [lean_vol_vix], QLIKE-selected	0.0970	0.0948	−11.1	−3.10	0.871	2
tuned per-bar XGBoost [lean_vol_vix], QLIKE-selected	0.1049	0.1027	−3.9	−0.86	0.912	0
tuned per-bar XGBoost [lean_vol_vix], MSE-selected	0.1041	0.1015	−4.7	−1.04	0.926	0
tuned per-bar random forest [all_features], QLIKE-selected	0.1029	0.1003	−5.8	−1.67	0.797	1
tuned per-bar random forest [all_features], MSE-selected	0.1085	0.1059	−0.6	−0.13	0.707	1
tuned per-bar XGBoost [baseline], MSE-selected	0.1100	0.1069	+0.7	+0.12	1.076	0
tuned per-bar LightGBM [baseline], MSE-selected	0.1071	0.1043	−1.9	−0.34	1.029	0
tuned per-bar LightGBM [baseline], QLIKE-selected	0.1027	0.1004	−5.9	−1.16	1.050	0
tuned per-bar XGBoost [baseline], QLIKE-selected	0.1055	0.1030	−3.4	−0.56	1.077	0
tuned per-bar random forest [baseline], QLIKE-selected	0.0989	0.0965	−9.4	−1.86	1.042	0
tuned per-bar random forest [baseline], MSE-selected	0.1089	0.1057	−0.3	−0.05	1.049	0
per-bar tree (random search, daily)						
tuned per-bar XGBoost, daily refit [lean_vol_vix], MSE-selected	0.1031	0.1006	−5.6	−1.27	0.927	0
tuned per-bar random forest, daily refit [lean_vol_vix], MSE-selected	0.0974	0.0951	−10.8	−2.88	0.885	1
tuned per-bar LightGBM, daily refit [lean_vol_vix], QLIKE-selected	0.0982	0.0962	−10.1	−2.37	0.912	0
tuned per-bar XGBoost, daily refit [all_features], MSE-selected	0.1009	0.0984	−7.6	−1.77	0.858	0
tuned per-bar XGBoost, daily refit [all_features], QLIKE-selected	0.1009	0.0985	−7.6	−1.76	0.859	0
tuned per-bar LightGBM, daily refit [all_features], MSE-selected	0.0990	0.0966	−9.3	−2.16	0.903	0
tuned per-bar random forest, daily refit [lean_vol_vix], QLIKE-selected	0.0949	0.0926	−13.1	−3.59	0.883	2
tuned per-bar XGBoost, daily refit [lean_vol_vix], QLIKE-selected	0.1029	0.1006	−5.8	−1.29	0.924	0
tuned per-bar LightGBM, daily refit [all_features], QLIKE-selected	0.0980	0.0956	−10.2	−2.47	0.880	0
tuned per-bar LightGBM, daily refit [lean_vol_vix], MSE-selected	0.1041	0.1015	−4.7	−1.12	0.914	0
tuned per-bar random forest, daily refit [all_features], MSE-selected	0.1045	0.1020	−4.3	−1.13	0.773	1
tuned per-bar random forest, daily refit [all_features], QLIKE-selected	0.1002	0.0976	−8.2	−2.30	0.834	1
tuned per-bar XGBoost, daily refit [baseline], QLIKE-selected	0.1052	0.1028	−3.6	−0.60	1.074	0
tuned per-bar LightGBM, daily refit [baseline], QLIKE-selected	0.1026	0.1002	−6.0	−1.19	1.039	0
tuned per-bar random forest, daily refit [baseline], MSE-selected	0.1090	0.1058	−0.1	−0.02	1.044	0
tuned per-bar LightGBM, daily refit [baseline], MSE-selected	0.1073	0.1045	−1.7	−0.30	1.028	0
tuned per-bar XGBoost, daily refit [baseline], MSE-selected	0.1098	0.1066	+0.6	+0.10	1.071	0
tuned per-bar random forest, daily refit [baseline], QLIKE-selected	0.0995	0.0971	−8.8	−1.72	1.035	0
per-bar tree (Optuna, TUNE_PER=1, best-of-50)						
Optuna per-bar random forest, TUNE_PER 1, best of 50 [lean_vol_vix]	0.0966	0.0941	−11.5	−3.15	0.902	1
Optuna per-bar XGBoost, TUNE_PER 1, best of 50 [lean_vol_vix]	0.1020	0.0995	−6.6	−1.82	0.877	0
Optuna per-bar LightGBM, TUNE_PER 1, best of 50 [lean_vol_vix]	0.1002	0.0978	−8.2	−2.03	0.919	1
Optuna per-bar LightGBM, TUNE_PER 1, best of 50 [all_features]	0.1032	0.1010	−5.5	−1.43	0.883	1
Optuna per-bar XGBoost, TUNE_PER 1, best of 50 [all_features]	0.1017	0.0993	−6.9	−1.96	0.865	1
Optuna per-bar LightGBM, TUNE_PER 1, best of 50 [baseline]	0.1055	0.1028	−3.4	−0.67	1.028	0
Optuna per-bar XGBoost, TUNE_PER 1, best of 50 [baseline]	0.1036	0.1011	−5.1	−0.99	0.992	0
Optuna per-bar random forest, TUNE_PER 1, best of 50 [baseline]	0.1015	0.0990	−7.1	−1.49	1.042	0
Optuna per-bar random forest, TUNE_PER 1, best of 50 [all_features]	0.1012	0.0985	−7.3	−2.01	0.868	1
per-bar tree (Optuna, TUNE_PER=5, best-of-50)						
Optuna per-bar random forest, TUNE_PER 5, best of 50 [lean_vol_vix]	0.0964	0.0939	−11.7	−3.25	0.887	1
Optuna per-bar LightGBM, TUNE_PER 5, best of 50 [lean_vol_vix]	0.0997	0.0973	−8.7	−2.22	0.887	0
Optuna per-bar XGBoost, TUNE_PER 5, best of 50 [all_features]	0.1005	0.0982	−7.9	−2.11	0.820	1
Optuna per-bar XGBoost, TUNE_PER 5, best of 50 [lean_vol_vix]	0.0999	0.0975	−8.5	−2.19	0.899	1

(continued)

Forecast	QLIKE	QLIKE, trimmed	% vs R	DM	$\overline{RV}/\overline{F}$	extreme (all sess.)
Optuna per-bar LightGBM, TUNE_PER 5, best of 50 [all_features]	0.1014	0.0994	−7.1	−1.90	0.892	0
Optuna per-bar random forest, TUNE_PER 5, best of 50 [all_features]	0.1012	0.0985	−7.3	−2.04	0.863	1
Optuna per-bar XGBoost, TUNE_PER 5, best of 50 [baseline]	0.1023	0.0999	−6.3	−1.16	0.993	0
Optuna per-bar LightGBM, TUNE_PER 5, best of 50 [baseline]	0.1044	0.1016	−4.4	−0.84	1.025	0
Optuna per-bar random forest, TUNE_PER 5, best of 50 [baseline]	0.1015	0.0990	−7.1	−1.44	1.036	0
per-bar tree (Optuna, TUNE_PER=25, best-of-50)						
Optuna per-bar LightGBM, TUNE_PER 25, best of 50 [all_features]	0.1064	0.1042	−2.5	−0.60	0.793	1
Optuna per-bar XGBoost, TUNE_PER 25, best of 50, QLIKE-selected [all_features]	0.1039	0.1016	−4.9	−1.09	0.749	0
Optuna per-bar LightGBM, TUNE_PER 25, best of 50, QLIKE-selected [lean_vol_vix]	0.0988	0.0968	−9.5	−2.25	0.857	0
Optuna per-bar XGBoost, TUNE_PER 25, best of 50, QLIKE-selected [lean_vol_vix]	0.0968	0.0943	−11.4	−2.77	0.962	1
Optuna per-bar LightGBM, TUNE_PER 25, best of 50 [lean_vol_vix]	0.0991	0.0968	−9.3	−1.98	0.921	0
Optuna per-bar random forest, TUNE_PER 25, best of 50 [lean_vol_vix]	0.0989	0.0964	−9.5	−2.59	0.872	1
Optuna per-bar random forest, TUNE_PER 25, best of 50, QLIKE-selected [lean_vol_vix]	0.0958	0.0934	−12.3	−3.36	0.885	1
Optuna per-bar XGBoost, TUNE_PER 25, best of 50 [all_features]	0.1036	0.1015	−5.1	−1.16	0.775	0
Optuna per-bar LightGBM, TUNE_PER 25, best of 50, QLIKE-selected [all_features]	0.1045	0.1020	−4.3	−0.98	0.849	0
Optuna per-bar random forest, TUNE_PER 25, best of 50, QLIKE-selected [all_features]	0.1023	0.0997	−6.3	−1.85	0.816	1
Optuna per-bar random forest, TUNE_PER 25, best of 50 [all_features]	0.1038	0.1009	−5.0	−1.46	0.819	1
Optuna per-bar XGBoost, TUNE_PER 25, best of 50 [lean_vol_vix]	0.1050	0.1026	−3.9	−0.94	0.877	1
Optuna per-bar LightGBM, TUNE_PER 25, best of 50 [baseline]	0.1052	0.1026	−3.7	−0.68	1.040	0
Optuna per-bar XGBoost, TUNE_PER 25, best of 50 [baseline]	0.1032	0.1007	−5.5	−1.01	1.035	0
Optuna per-bar XGBoost, TUNE_PER 25, best of 50, QLIKE-selected [baseline]	0.1034	0.1011	−5.3	−0.90	1.007	0
Optuna per-bar LightGBM, TUNE_PER 25, best of 50, QLIKE-selected [baseline]	0.1036	0.1007	−5.1	−0.98	1.046	0
Optuna per-bar random forest, TUNE_PER 25, best of 50, QLIKE-selected [baseline]	0.1008	0.0983	−7.7	−1.50	1.020	0
Optuna per-bar random forest, TUNE_PER 25, best of 50 [baseline]	0.1017	0.0992	−6.8	−1.38	1.031	0
per-bar tree (Optuna, TUNE_PER=250, best-of-50)						
Optuna per-bar LightGBM, TUNE_PER 250, best of 50 [lean_vol_vix]	0.1035	0.1010	−5.2	−1.33	0.895	0
Optuna per-bar LightGBM, TUNE_PER 250, best of 50 [all_features]	0.1006	0.0983	−7.8	−1.72	0.908	0
Optuna per-bar XGBoost, TUNE_PER 250, best of 50 [lean_vol_vix]	0.1064	0.1040	−2.5	−0.59	0.926	0
Optuna per-bar random forest, TUNE_PER 250, best of 50 [lean_vol_vix]	0.0953	0.0930	−12.7	−3.43	0.886	1
Optuna per-bar XGBoost, TUNE_PER 250, best of 50 [all_features]	0.1013	0.0991	−7.3	−1.47	0.857	0
Optuna per-bar XGBoost, TUNE_PER 250, best of 50 [baseline]	0.1092	0.1062	+0.0	+0.00	1.033	0
Optuna per-bar LightGBM, TUNE_PER 250, best of 50 [baseline]	0.1100	0.1068	+0.7	+0.12	1.039	0
Optuna per-bar random forest, TUNE_PER 250, best of 50 [all_features]	0.1027	0.1001	−5.9	−1.60	0.826	1
Optuna per-bar random forest, TUNE_PER 250, best of 50 [baseline]	0.1060	0.1036	−2.9	−0.51	1.055	0
per-bar tree (Optuna, TUNE_PER=1, best-of-25)						
Optuna per-bar LightGBM, TUNE_PER 1, best of 25 [all_features]	0.1028	0.1004	−5.8	−1.58	0.878	1
Optuna per-bar LightGBM, TUNE_PER 1, best of 25 [lean_vol_vix]	0.0995	0.0971	−8.9	−2.23	0.908	0
Optuna per-bar XGBoost, TUNE_PER 1, best of 25 [all_features]	0.1000	0.0976	−8.4	−2.33	0.865	1
Optuna per-bar random forest, TUNE_PER 1, best of 25 [lean_vol_vix]	0.0969	0.0944	−11.3	−3.16	0.895	1
Optuna per-bar XGBoost, TUNE_PER 1, best of 25 [baseline]	0.1037	0.1011	−5.1	−0.96	0.991	0
Optuna per-bar XGBoost, TUNE_PER 1, best of 25 [lean_vol_vix]	0.1002	0.0978	−8.2	−2.36	0.887	0
Optuna per-bar random forest, TUNE_PER 1, best of 25 [all_features]	0.1006	0.0979	−7.9	−2.13	0.873	1
Optuna per-bar random forest, TUNE_PER 1, best of 25 [baseline]	0.1013	0.0988	−7.3	−1.52	1.043	0
Optuna per-bar LightGBM, TUNE_PER 1, best of 25 [baseline]	0.1058	0.1029	−3.1	−0.66	1.032	0
per-bar tree (Optuna, TUNE_PER=25, best-of-25)						
Optuna per-bar random forest, TUNE_PER 25, best of 25 [lean_vol_vix]	0.0974	0.0948	−10.8	−3.02	0.867	2
Optuna per-bar XGBoost, TUNE_PER 25, best of 25 [all_features]	0.1031	0.1008	−5.6	−1.31	0.743	0
Optuna per-bar LightGBM, TUNE_PER 25, best of 25 [lean_vol_vix]	0.1003	0.0979	−8.2	−1.90	0.889	1
Optuna per-bar XGBoost, TUNE_PER 25, best of 25 [lean_vol_vix]	0.1034	0.1010	−5.3	−1.32	0.876	0
Optuna per-bar LightGBM, TUNE_PER 25, best of 25 [all_features]	0.1026	0.1003	−6.1	−1.43	0.807	0
Optuna per-bar random forest, TUNE_PER 25, best of 25 [all_features]	0.1022	0.0996	−6.4	−1.79	0.820	1
Optuna per-bar XGBoost, TUNE_PER 25, best of 25 [baseline]	0.1035	0.1009	−5.2	−0.93	1.018	0
Optuna per-bar LightGBM, TUNE_PER 25, best of 25 [baseline]	0.1040	0.1016	−4.8	−0.91	1.039	0
Optuna per-bar random forest, TUNE_PER 25, best of 25 [baseline]	0.1015	0.0990	−7.0	−1.41	1.031	0
per-bar tree (Optuna, TUNE_PER=1, best-of-10)						
Optuna per-bar LightGBM, TUNE_PER 1, best of 10 [lean_vol_vix]	0.0996	0.0974	−8.8	−2.34	0.885	0
Optuna per-bar XGBoost, TUNE_PER 1, best of 10 [all_features]	0.1031	0.1007	−5.6	−1.35	0.840	1
Optuna per-bar XGBoost, TUNE_PER 1, best of 10 [lean_vol_vix]	0.0998	0.0973	−8.6	−2.22	0.859	0
Optuna per-bar random forest, TUNE_PER 1, best of 10 [all_features]	0.1008	0.0981	−7.7	−2.13	0.855	1
Optuna per-bar LightGBM, TUNE_PER 1, best of 10 [all_features]	0.1031	0.1006	−5.6	−1.53	0.846	0
Optuna per-bar random forest, TUNE_PER 1, best of 10 [lean_vol_vix]	0.0975	0.0948	−10.7	−2.99	0.884	2
Optuna per-bar random forest, TUNE_PER 1, best of 10 [baseline]	0.1014	0.0989	−7.1	−1.48	1.035	0
Optuna per-bar LightGBM, TUNE_PER 1, best of 10 [baseline]	0.1051	0.1025	−3.8	−0.77	1.013	0
Optuna per-bar XGBoost, TUNE_PER 1, best of 10 [baseline]	0.1024	0.0999	−6.2	−1.24	1.002	0

(continued)

Forecast	QLIKE	QLIKE, trimmed	% vs R	DM	$\overline{RV}/\overline{F}$	extreme (all sess.)
per-bar tree (Optuna, TUNE_PER=25, best-of-10)						
Optuna per-bar XGBoost, TUNE_PER 25, best of 10 [`lean_vol_vix`]	0.1029	0.1005	−5.8	−1.54	0.780	0
Optuna per-bar LightGBM, TUNE_PER 25, best of 10 [`lean_vol_vix`]	0.0982	0.0959	−10.1	−2.53	0.845	1
Optuna per-bar random forest, TUNE_PER 25, best of 10 [`all_features`]	0.1006	0.0979	−7.9	−2.27	0.831	1
Optuna per-bar LightGBM, TUNE_PER 25, best of 10 [`all_features`]	0.1055	0.1031	−3.4	−0.76	0.855	0
Optuna per-bar XGBoost, TUNE_PER 25, best of 10 [`all_features`]	0.1076	0.1053	−1.4	−0.34	0.739	0
Optuna per-bar random forest, TUNE_PER 25, best of 10 [`lean_vol_vix`]	0.0962	0.0937	−11.9	−3.21	0.885	2
Optuna per-bar LightGBM, TUNE_PER 25, best of 10 [`baseline`]	0.1058	0.1032	−3.1	−0.61	1.060	0
Optuna per-bar XGBoost, TUNE_PER 25, best of 10 [`baseline`]	0.1030	0.1003	−5.7	−1.06	1.023	0
Optuna per-bar random forest, TUNE_PER 25, best of 10 [`baseline`]	0.1020	0.0995	−6.6	−1.26	1.021	0
implied-vol representations						
per-bar elastic net [`ivrep_term`]	0.0966	0.0943	−11.5	−3.74	0.972	3
per-bar ridge [`ivrep_term`]	0.1166	0.1145	+6.8	+0.44	0.981	4
per-bar ridge [`ivrep_vrp`]	0.1102	0.1080	+0.9	+0.07	0.934	5
per-bar lasso [`ivrep_vrp`]	0.0947	0.0921	−13.2	−3.43	0.923	5
per-bar lasso [`ivrep_term`]	0.0954	0.0935	−12.6	−3.72	0.964	4
per-bar ridge [`ivrep_all`]	0.1113	0.1091	+1.9	+0.17	0.982	4
per-bar elastic net [`ivrep_vrp`]	0.0941	0.0919	−13.8	−4.11	0.928	3
per-bar ridge [`ivrep_target_scale`]	0.1124	0.1102	+2.9	+0.22	0.919	4
per-bar lasso [`ivrep_innovations`]	0.0983	0.0957	−10.0	−2.66	0.995	3
per-bar lasso [`ivrep_target_scale`]	0.0950	0.0932	−13.0	−3.17	0.870	2
per-bar ridge [`ivrep_innovations`]	0.1133	0.1113	+3.7	+0.27	1.007	4
per-bar lasso [`ivrep_all`]	0.0996	0.0970	−8.8	−2.24	0.918	3
per-bar elastic net [`ivrep_all`]	0.0953	0.0932	−12.7	−3.07	0.926	3
per-bar elastic net [`ivrep_innovations`]	0.0965	0.0944	−11.6	−2.79	1.009	3
per-bar elastic net [`ivrep_target_scale`]	0.0965	0.0946	−11.6	−2.94	0.875	4
Table B: scored on the forecast's own days, paired with the reference on them (550 trade days from 2022-01-03)						
chain-period buckets						
per-bar elastic net [`ivrep_slice_slope`], 500 sessions	0.0915	0.0890	−4.1	−0.94	0.972	6
per-bar ridge [`ivrep_slice_slope`], 500 sessions	0.0895	0.0872	−6.2	−1.44	0.952	5
per-bar lasso [`ivrep_slice_slope`], 500 sessions	0.0957	0.0926	+0.3	+0.06	0.986	5
per-bar ridge [`ivslice_only`], 500 sessions	0.0944	0.0920	−1.0	−0.23	0.981	0
per-bar elastic net [`ivslice_only`], 500 sessions	0.0970	0.0947	+1.6	+0.32	0.998	0
per-bar lasso [`ivslice_only`], 500 sessions	0.0993	0.0977	+4.0	+0.53	1.003	0
per-bar lasso [`mfiv_only`], 500 sessions	0.1030	0.1011	+7.9	+1.25	1.005	0
per-bar elastic net [`mfiv_only`], 500 sessions	0.1020	0.1001	+6.8	+1.10	1.001	0
per-bar ridge [`mfiv_only`], 500 sessions	0.1017	0.0996	+6.6	+1.12	1.014	0

Table 2: The last-30-min trade: sign(s) on the straddle (nearest out-of-the-money call and put, same-day expiry, one position), entered at 15:30, held to the close; buy when the plain 16:00 forecast f^2B exceeds the 15:30 implied variance, sell otherwise. Returns per unit of premium; Sharpe annualized by $\sqrt{252}$ per trade day; $^{\times}$: the entry pays the spread (buy at the ask, sell at the bid); hit: share of days with a positive midpoint return; buys: days the rule buys. ΔSharpe: the row minus the reference R (or minus always short) on the same days, point estimates; t vs R: Newey–West HAC t statistic of the mean daily midpoint-return difference against R.

Forecast	Sharpe	Sharpe$^{\times}$	mean	mean$^{\times}$	hit %	buys	ΔSharpe vs R	ΔSharpe$^{\times}$ vs R	t vs R	ΔSharpe vs short	ΔSharpe$^{\times}$ vs short
Table A: the same 866 trade days, 2020-01-03 to 2024-04-30											
rule											
always short (no forecast)	0.20	−0.27	+0.014	−0.020	61.8	0	−1.08	−1.07	−1.71		
per-bar linear											
per-bar ridge [`lean_vol_vix`]	2.11	1.63	+0.148	+0.115	57.7	257	+0.83	+0.83	+1.87	+1.90	+1.90
per-bar elastic net [`all_features`]	2.04	1.56	+0.144	+0.110	58.1	256	+0.76	+0.76	+1.63	+1.84	+1.84
per-bar lasso [`all_features`]	2.00	1.53	+0.141	+0.107	58.0	259	+0.72	+0.73	+1.53	+1.80	+1.80
per-bar elastic net [`lean_vol_vix`], pre-tuned	1.93	1.45	+0.136	+0.102	57.9	250	+0.65	+0.65	+1.34	+1.73	+1.72
per-bar elastic net [`all_features`], pre-tuned	1.92	1.43	+0.135	+0.102	57.7	245	+0.64	+0.63	+1.16	+1.72	+1.71
per-bar ridge [`lean_vol_vix`], pre-tuned	1.88	1.40	+0.133	+0.099	57.2	256	+0.60	+0.60	+1.22	+1.68	+1.67
per-bar lasso [`all_features`], pre-tuned	1.88	1.40	+0.132	+0.099	57.6	256	+0.60	+0.60	+1.29	+1.67	+1.67
per-bar lasso [`lean_vol_vix`]	1.73	1.25	+0.122	+0.088	57.7	247	+0.45	+0.45	+0.98	+1.53	+1.52
per-bar ridge [`all_features`]	1.71	1.22	+0.121	+0.087	57.9	240	+0.43	+0.43	+0.81	+1.50	+1.50
per-bar elastic net [`lean_vol_vix`]	1.62	1.14	+0.114	+0.081	57.0	259	+0.34	+0.34	+0.77	+1.41	+1.41
per-bar lasso [`lean_vol_vix`], pre-tuned	1.44	0.96	+0.102	+0.068	56.8	251	+0.16	+0.16	+0.29	+1.24	+1.23

(continued)

Forecast	Sharpe	Sharpe$^{\times}$	mean	mean$^{\times}$	hit %	buys	ΔSharpe vs R	ΔSharpe$^{\times}$ vs R	t vs R	ΔSharpe vs short	ΔSharpe$^{\times}$ vs short
per-bar ridge [`all_features`], pre-tuned	1.42	0.93	+0.100	+0.067	57.5	235	+0.14	+0.13	+0.28	+1.21	+1.21
per-bar elastic net [`baseline`], pre-tuned	1.37	0.89	+0.097	+0.063	56.7	242	+0.09	+0.09	+0.16	+1.17	+1.16
per-bar lasso [`baseline`], pre-tuned	1.37	0.89	+0.097	+0.063	56.7	242	+0.09	+0.09	+0.16	+1.17	+1.16
per-bar ridge [`baseline`], pre-tuned	1.37	0.89	+0.097	+0.063	56.7	242	+0.09	+0.09	+0.16	+1.17	+1.16
per-bar OLS [`baseline`]	1.37	0.89	+0.097	+0.063	56.7	242	+0.09	+0.09	+0.16	+1.17	+1.16
per-bar ridge [`baseline`]	1.36	0.88	+0.097	+0.063	56.5	250	+0.08	+0.08	+0.15	+1.16	+1.16
per-bar lasso [`baseline`]	1.29	0.81	+0.091	+0.057	56.4	247	+0.01	+0.01	+0.01	+1.09	+1.08
per-bar elastic net [`baseline`]	1.23	0.74	+0.087	+0.053	56.2	244	−0.06	−0.06	−0.10	+1.02	+1.02
VIX-only family											
per-bar ridge [`live_vix_only`]	2.53	2.05	+0.177	+0.144	58.8	272	+1.24	+1.25	+2.78	+2.32	+2.32
per-bar ridge [`free_vix_only`]	2.49	2.02	+0.175	+0.141	58.9	271	+1.21	+1.22	+2.65	+2.29	+2.29
per-bar ridge [`free_feasible`]	2.08	1.59	+0.146	+0.113	58.3	254	+0.79	+0.79	+1.47	+1.87	+1.86
per-bar ridge [`live_vix_rvol`]	2.04	1.56	+0.143	+0.110	57.4	272	+0.76	+0.76	+1.66	+1.83	+1.83
per-bar ridge [`free_feasible_vol`]	2.03	1.55	+0.143	+0.110	57.6	268	+0.75	+0.75	+1.51	+1.83	+1.83
per-bar lasso [`free_vix_only`]	1.91	1.43	+0.135	+0.101	58.2	251	+0.63	+0.63	+1.30	+1.71	+1.70
per-bar lasso [`free_feasible_vol`]	1.81	1.33	+0.128	+0.094	57.2	252	+0.53	+0.53	+1.11	+1.60	+1.60
per-bar elastic net [`free_vix_only`]	1.70	1.22	+0.120	+0.087	57.4	256	+0.42	+0.43	+0.98	+1.50	+1.50
per-bar elastic net [`live_vix_rvol`]	1.64	1.16	+0.116	+0.082	56.9	258	+0.36	+0.36	+0.78	+1.43	+1.43
per-bar elastic net [`free_feasible_vol`]	1.60	1.13	+0.113	+0.080	56.8	261	+0.32	+0.33	+0.76	+1.40	+1.40
per-bar lasso [`live_vix_only`]	1.58	1.09	+0.112	+0.078	58.0	245	+0.30	+0.29	+0.51	+1.38	+1.36
per-bar lasso [`live_vix_rvol`]	1.55	1.07	+0.110	+0.076	56.8	255	+0.27	+0.27	+0.61	+1.35	+1.35
per-bar elastic net [`live_vix_only`]	1.53	1.04	+0.109	+0.075	57.7	249	+0.25	+0.24	+0.46	+1.33	+1.32
per-bar lasso [`free_feasible`]	1.40	0.92	+0.099	+0.066	56.9	250	+0.12	+0.12	+0.31	+1.20	+1.20
per-bar elastic net [`free_feasible`]	1.33	0.85	+0.094	+0.060	56.0	254	+0.05	+0.05	+0.11	+1.12	+1.12
per-bar lasso [`vix_only`]	1.13	0.64	+0.080	+0.046	57.0	203	−0.15	−0.16	−0.27	+0.92	+0.92
per-bar ridge [`vix_only`]	1.09	0.61	+0.077	+0.043	56.5	214	−0.19	−0.19	−0.38	+0.89	+0.88
per-bar ridge [`vix_rvol`]	0.91	0.43	+0.065	+0.031	55.9	213	−0.37	−0.37	−0.77	+0.71	+0.71
per-bar lasso [`vix_rvol`]	0.79	0.31	+0.056	+0.022	56.5	208	−0.49	−0.49	−0.98	+0.59	+0.58
per-bar elastic net [`vix_only`]	0.79	0.31	+0.056	+0.022	55.4	229	−0.50	−0.49	−1.03	+0.58	+0.58
per-bar elastic net [`vix_rvol`]	0.74	0.26	+0.053	+0.019	55.4	231	−0.54	−0.53	−1.17	+0.54	+0.54
per-bar tree (untuned)											
per-bar XGBoost [`lean_vol_vix`]	1.78	1.30	+0.126	+0.092	57.7	253	+0.50	+0.50	+1.00	+1.58	+1.57
per-bar random forest (dev-tuned) [`all_features`]	1.75	1.27	+0.123	+0.090	56.0	288	+0.46	+0.47	+0.90	+1.54	+1.55
per-bar LightGBM [`lean_vol_vix`]	1.68	1.20	+0.119	+0.085	56.8	263	+0.40	+0.40	+0.75	+1.48	+1.48
per-bar random forest [`lean_vol_vix`]	1.60	1.11	+0.113	+0.079	56.9	264	+0.32	+0.31	+0.58	+1.40	+1.39
per-bar XGBoost [`all_features`]	1.55	1.07	+0.109	+0.076	56.7	258	+0.27	+0.27	+0.45	+1.34	+1.34
per-bar LightGBM [`all_features`]	1.37	0.88	+0.097	+0.063	56.8	247	+0.09	+0.09	+0.15	+1.16	+1.16
per-bar random forest (dev-tuned) [`lean_vol_vix`]	1.30	0.82	+0.092	+0.058	56.1	271	+0.02	+0.02	+0.04	+1.10	+1.09
per-bar random forest [`all_features`]	1.27	0.80	+0.090	+0.057	54.6	292	−0.01	+0.00	−0.02	+1.07	+1.07
per-bar XGBoost [`baseline`]	1.20	0.72	+0.085	+0.051	55.9	265	−0.08	−0.07	−0.17	+1.00	+1.00
per-bar random forest [`baseline`]	0.98	0.50	+0.069	+0.036	55.2	263	−0.30	−0.30	−0.65	+0.77	+0.77
per-bar random forest (dev-tuned) [`baseline`]	0.75	0.27	+0.053	+0.020	55.5	264	−0.53	−0.53	−0.90	+0.55	+0.55
per-bar LightGBM [`baseline`]	0.73	0.25	+0.052	+0.018	55.8	264	−0.55	−0.55	−0.95	+0.53	+0.52
per-bar tree (untuned, daily refit)											
per-bar XGBoost, daily refit [`lean_vol_vix`]	1.76	1.28	+0.124	+0.091	57.9	244	+0.48	+0.48	+0.91	+1.56	+1.55
per-bar random forest, daily refit [`lean_vol_vix`]	1.58	1.09	+0.112	+0.078	57.2	256	+0.30	+0.29	+0.54	+1.38	+1.36
per-bar random forest (dev-tuned), daily refit [`lean_vol_vix`]	1.40	0.90	+0.099	+0.065	57.3	261	+0.11	+0.10	+0.19	+1.19	+1.18
per-bar LightGBM, daily refit [`all_features`]	1.33	0.84	+0.094	+0.061	57.4	250	+0.05	+0.04	+0.08	+1.13	+1.12
per-bar LightGBM, daily refit [`lean_vol_vix`]	1.26	0.77	+0.089	+0.056	57.5	243	−0.02	−0.03	−0.03	+1.06	+1.04
per-bar XGBoost, daily refit [`all_features`]	1.12	0.63	+0.079	+0.045	56.7	250	−0.16	−0.17	−0.25	+0.91	+0.90
per-bar random forest (dev-tuned), daily refit [`baseline`]	1.07	0.59	+0.076	+0.042	56.2	258	−0.21	−0.21	−0.36	+0.87	+0.86
per-bar XGBoost, daily refit [`baseline`]	0.95	0.48	+0.068	+0.034	55.2	265	−0.33	−0.32	−0.73	+0.75	+0.75
per-bar random forest (dev-tuned), daily refit [`all_features`]	0.85	0.37	+0.060	+0.027	55.2	267	−0.43	−0.43	−0.70	+0.65	+0.64
per-bar random forest, daily refit [`all_features`]	0.85	0.37	+0.060	+0.026	54.5	277	−0.44	−0.43	−0.88	+0.64	+0.64
per-bar LightGBM, daily refit [`baseline`]	0.79	0.31	+0.056	+0.022	56.1	261	−0.49	−0.49	−0.81	+0.59	+0.59
per-bar random forest, daily refit [`baseline`]	0.66	0.19	+0.047	+0.013	55.3	264	−0.62	−0.61	−1.05	+0.46	+0.46
per-bar tree (tuned)											
tuned per-bar random forest [`lean_vol_vix`], MSE-selected	1.81	1.34	+0.128	+0.094	57.3	267	+0.53	+0.54	+1.26	+1.61	+1.61
tuned per-bar LightGBM [`all_features`], MSE-selected	1.81	1.33	+0.128	+0.094	56.9	264	+0.53	+0.54	+1.04	+1.61	+1.61
tuned per-bar LightGBM [`lean_vol_vix`], MSE-selected	1.78	1.31	+0.126	+0.092	56.0	290	+0.50	+0.51	+1.05	+1.58	+1.58
tuned per-bar LightGBM [`lean_vol_vix`], QLIKE-selected	1.72	1.25	+0.122	+0.088	57.4	262	+0.44	+0.45	+1.09	+1.52	+1.52
tuned per-bar XGBoost [`all_features`], MSE-selected	1.68	1.19	+0.118	+0.085	57.9	250	+0.40	+0.39	+0.66	+1.47	+1.46
tuned per-bar XGBoost [`all_features`], QLIKE-selected	1.61	1.12	+0.114	+0.080	57.7	249	+0.33	+0.32	+0.54	+1.41	+1.40
tuned per-bar LightGBM [`all_features`], QLIKE-selected	1.54	1.06	+0.109	+0.075	56.6	269	+0.26	+0.26	+0.48	+1.33	+1.33

(continued)

Forecast	Sharpe	Sharpe$^{\times}$	mean	mean$^{\times}$	hit %	buys	ΔSharpe vs R	ΔSharpe$^{\times}$ vs R	t vs R	ΔSharpe vs short	ΔSharpe$^{\times}$ vs short
tuned per-bar random forest [`lean_vol_vix`], QLIKE-selected	1.49	1.00	+0.105	+0.071	56.7	262	+0.20	+0.20	+0.37	+1.28	+1.27
tuned per-bar XGBoost [`lean_vol_vix`], QLIKE-selected	1.35	0.88	+0.096	+0.062	55.2	269	+0.07	+0.08	+0.14	+1.15	+1.15
tuned per-bar XGBoost [`lean_vol_vix`], MSE-selected	1.24	0.76	+0.088	+0.054	55.0	255	−0.04	−0.04	−0.06	+1.04	+1.04
tuned per-bar random forest [`all_features`], QLIKE-selected	1.24	0.76	+0.088	+0.054	55.2	273	−0.04	−0.03	−0.08	+1.03	+1.04
tuned per-bar random forest [`all_features`], MSE-selected	1.12	0.64	+0.079	+0.046	55.5	262	−0.16	−0.16	−0.30	+0.91	+0.92
tuned per-bar XGBoost [`baseline`], MSE-selected	0.99	0.51	+0.070	+0.036	53.7	290	−0.29	−0.29	−0.62	+0.78	+0.79
tuned per-bar LightGBM [`baseline`], MSE-selected	0.95	0.48	+0.068	+0.034	54.8	278	−0.33	−0.32	−0.66	+0.75	+0.75
tuned per-bar LightGBM [`baseline`], QLIKE-selected	0.94	0.46	+0.067	+0.033	54.5	279	−0.34	−0.34	−0.71	+0.74	+0.74
tuned per-bar XGBoost [`baseline`], QLIKE-selected	0.92	0.44	+0.066	+0.032	55.7	265	−0.36	−0.36	−0.60	+0.72	+0.72
tuned per-bar random forest [`baseline`], QLIKE-selected	0.78	0.30	+0.055	+0.021	55.7	269	−0.50	−0.50	−0.84	+0.57	+0.57
tuned per-bar random forest [`baseline`], MSE-selected	0.65	0.17	+0.046	+0.012	54.7	285	−0.63	−0.63	−1.06	+0.44	+0.44
per-bar tree (random search, daily)											
tuned per-bar XGBoost, daily refit [`lean_vol_vix`], MSE-selected	1.95	1.47	+0.137	+0.104	57.5	257	+0.67	+0.67	+1.37	+1.74	+1.75
tuned per-bar random forest, daily refit [`lean_vol_vix`], MSE-selected	1.86	1.38	+0.131	+0.097	57.2	268	+0.57	+0.58	+1.37	+1.65	+1.65
tuned per-bar LightGBM, daily refit [`lean_vol_vix`], QLIKE-selected	1.78	1.30	+0.126	+0.092	57.6	250	+0.50	+0.50	+1.23	+1.58	+1.58
tuned per-bar XGBoost, daily refit [`all_features`], MSE-selected	1.77	1.27	+0.125	+0.091	58.7	241	+0.48	+0.48	+0.82	+1.56	+1.55
tuned per-bar XGBoost, daily refit [`all_features`], QLIKE-selected	1.76	1.27	+0.124	+0.091	58.7	241	+0.48	+0.47	+0.82	+1.56	+1.55
tuned per-bar LightGBM, daily refit [`all_features`], MSE-selected	1.61	1.12	+0.114	+0.080	57.7	249	+0.33	+0.32	+0.59	+1.40	+1.40
tuned per-bar random forest, daily refit [`lean_vol_vix`], QLIKE-selected	1.47	0.98	+0.104	+0.070	57.3	261	+0.19	+0.18	+0.34	+1.26	+1.26
tuned per-bar XGBoost, daily refit [`lean_vol_vix`], QLIKE-selected	1.28	0.80	+0.091	+0.057	56.2	260	−0.00	−0.00	−0.00	+1.07	+1.07
tuned per-bar LightGBM, daily refit [`all_features`], QLIKE-selected	1.28	0.79	+0.090	+0.057	57.5	255	−0.00	−0.01	−0.01	+1.07	+1.06
tuned per-bar LightGBM, daily refit [`lean_vol_vix`], MSE-selected	1.25	0.77	+0.089	+0.055	55.7	277	−0.03	−0.03	−0.05	+1.05	+1.05
tuned per-bar random forest, daily refit [`all_features`], MSE-selected	1.18	0.70	+0.083	+0.050	56.5	256	−0.10	−0.10	−0.20	+0.97	+0.97
tuned per-bar random forest, daily refit [`all_features`], QLIKE-selected	1.13	0.64	+0.080	+0.046	56.4	253	−0.15	−0.15	−0.30	+0.92	+0.92
tuned per-bar XGBoost, daily refit [`baseline`], QLIKE-selected	1.09	0.61	+0.077	+0.043	55.4	269	−0.20	−0.19	−0.38	+0.88	+0.88
tuned per-bar LightGBM, daily refit [`baseline`], QLIKE-selected	0.90	0.43	+0.064	+0.030	55.1	276	−0.38	−0.37	−0.79	+0.70	+0.70
tuned per-bar random forest, daily refit [`baseline`], MSE-selected	0.68	0.20	+0.048	+0.014	54.5	293	−0.61	−0.60	−0.95	+0.47	+0.47
tuned per-bar LightGBM, daily refit [`baseline`], MSE-selected	0.64	0.16	+0.045	+0.012	55.1	278	−0.64	−0.64	−1.08	+0.44	+0.44
tuned per-bar XGBoost, daily refit [`baseline`], MSE-selected	0.57	0.09	+0.041	+0.007	53.6	287	−0.71	−0.70	−1.48	+0.37	+0.37
tuned per-bar random forest, daily refit [`baseline`], QLIKE-selected	0.46	−0.02	+0.033	−0.001	54.5	273	−0.82	−0.82	−1.31	+0.25	+0.26
per-bar tree (Optuna, TUNE_PER=1, best-of-50)											
Optuna per-bar random forest, TUNE_PER 1, best of 50 [`lean_vol_vix`]	1.50	1.01	+0.106	+0.072	57.5	257	+0.22	+0.21	+0.41	+1.30	+1.29
Optuna per-bar XGBoost, TUNE_PER 1, best of 50 [`lean_vol_vix`]	1.37	0.88	+0.097	+0.063	57.4	250	+0.08	+0.08	+0.14	+1.16	+1.15
Optuna per-bar LightGBM, TUNE_PER 1, best of 50 [`lean_vol_vix`]	1.28	0.79	+0.091	+0.057	57.5	255	−0.00	−0.01	−0.00	+1.08	+1.07
Optuna per-bar LightGBM, TUNE_PER 1, best of 50 [`all_features`]	1.23	0.75	+0.087	+0.054	58.1	246	−0.05	−0.05	−0.07	+1.03	+1.02
Optuna per-bar XGBoost, TUNE_PER 1, best of 50 [`all_features`]	1.21	0.72	+0.085	+0.052	57.0	251	−0.08	−0.08	−0.12	+1.00	+0.99
Optuna per-bar LightGBM, TUNE_PER 1, best of 50 [`baseline`]	0.95	0.47	+0.067	+0.033	55.8	274	−0.33	−0.33	−0.54	+0.74	+0.74
Optuna per-bar XGBoost, TUNE_PER 1, best of 50 [`baseline`]	0.80	0.32	+0.057	+0.023	55.2	265	−0.48	−0.48	−0.81	+0.60	+0.60
Optuna per-bar random forest, TUNE_PER 1, best of 50 [`baseline`]	0.64	0.16	+0.045	+0.012	54.3	267	−0.64	−0.63	−1.33	+0.44	+0.44

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Forecast	Sharpe	Sharpe$^{\times}$	mean	mean$^{\times}$	hit %	buys	ΔSharpe vs R	ΔSharpe$^{\times}$ vs R	t vs R	ΔSharpe vs short	ΔSharpe$^{\times}$ vs short
Optuna per-bar random forest, TUNE_PER 1, best of 50 [all_features]	0.59	0.12	+0.042	+0.008	55.3	258	−0.69	−0.68	−1.01	+0.39	+0.39
per-bar tree (Optuna, TUNE_PER=5, best-of-50)											
Optuna per-bar random forest, TUNE_PER 5, best of 50 [lean_vol_vix]	1.48	0.99	+0.104	+0.071	57.5	257	+0.19	+0.19	+0.35	+1.27	+1.26
Optuna per-bar LightGBM, TUNE_PER 5, best of 50 [lean_vol_vix]	1.28	0.79	+0.091	+0.057	57.4	250	+0.00	−0.00	+0.01	+1.08	+1.07
Optuna per-bar XGBoost, TUNE_PER 5, best of 50 [all_features]	1.16	0.67	+0.082	+0.048	56.5	260	−0.13	−0.13	−0.21	+0.95	+0.95
Optuna per-bar XGBoost, TUNE_PER 5, best of 50 [lean_vol_vix]	0.99	0.51	+0.070	+0.037	55.3	250	−0.29	−0.29	−0.50	+0.79	+0.78
Optuna per-bar LightGBM, TUNE_PER 5, best of 50 [all_features]	0.96	0.48	+0.068	+0.034	57.4	240	−0.32	−0.32	−0.51	+0.76	+0.75
Optuna per-bar random forest, TUNE_PER 5, best of 50 [all_features]	0.86	0.38	+0.061	+0.027	55.9	257	−0.42	−0.42	−0.65	+0.66	+0.65
Optuna per-bar XGBoost, TUNE_PER 5, best of 50 [baseline]	0.80	0.32	+0.057	+0.023	55.3	272	−0.48	−0.48	−0.82	+0.60	+0.60
Optuna per-bar LightGBM, TUNE_PER 5, best of 50 [baseline]	0.47	−0.01	+0.033	−0.001	54.7	271	−0.82	−0.81	−1.41	+0.26	+0.26
Optuna per-bar random forest, TUNE_PER 5, best of 50 [baseline]	0.41	−0.07	+0.029	−0.005	54.4	266	−0.87	−0.87	−1.44	+0.20	+0.21
per-bar tree (Optuna, TUNE_PER=25, best-of-50)											
Optuna per-bar LightGBM, TUNE_PER 25, best of 50 [all_features]	1.78	1.30	+0.126	+0.092	57.7	257	+0.50	+0.50	+0.89	+1.58	+1.57
Optuna per-bar XGBoost, TUNE_PER 25, best of 50, QLIKE-selected [all_features]	1.52	1.03	+0.108	+0.074	56.8	259	+0.24	+0.23	+0.41	+1.32	+1.31
Optuna per-bar LightGBM, TUNE_PER 25, best of 50, QLIKE-selected [lean_vol_vix]	1.49	1.00	+0.105	+0.072	58.1	266	+0.21	+0.20	+0.33	+1.29	+1.27
Optuna per-bar XGBoost, TUNE_PER 25, best of 50, QLIKE-selected [lean_vol_vix]	1.27	0.79	+0.090	+0.056	56.1	243	−0.01	−0.01	−0.02	+1.07	+1.06
Optuna per-bar LightGBM, TUNE_PER 25, best of 50 [lean_vol_vix]	1.27	0.78	+0.090	+0.056	57.6	262	−0.01	−0.02	−0.01	+1.07	+1.06
Optuna per-bar random forest, TUNE_PER 25, best of 50 [lean_vol_vix]	1.26	0.78	+0.089	+0.056	56.5	268	−0.02	−0.02	−0.04	+1.06	+1.05
Optuna per-bar random forest, TUNE_PER 25, best of 50, QLIKE-selected [lean_vol_vix]	1.24	0.75	+0.088	+0.054	56.6	259	−0.04	−0.05	−0.08	+1.03	+1.03
Optuna per-bar XGBoost, TUNE_PER 25, best of 50 [all_features]	1.23	0.74	+0.087	+0.053	56.0	262	−0.06	−0.06	−0.09	+1.02	+1.01
Optuna per-bar LightGBM, TUNE_PER 25, best of 50, QLIKE-selected [all_features]	1.16	0.68	+0.082	+0.048	56.4	261	−0.12	−0.12	−0.25	+0.95	+0.95
Optuna per-bar random forest, TUNE_PER 25, best of 50, QLIKE-selected [all_features]	1.07	0.59	+0.076	+0.042	55.9	267	−0.21	−0.20	−0.37	+0.87	+0.87
Optuna per-bar random forest, TUNE_PER 25, best of 50 [all_features]	1.02	0.55	+0.073	+0.039	55.5	264	−0.26	−0.25	−0.44	+0.82	+0.82
Optuna per-bar XGBoost, TUNE_PER 25, best of 50 [lean_vol_vix]	0.85	0.37	+0.060	+0.027	55.3	264	−0.43	−0.43	−0.68	+0.65	+0.65
Optuna per-bar LightGBM, TUNE_PER 25, best of 50 [baseline]	0.82	0.34	+0.058	+0.025	55.5	266	−0.46	−0.46	−0.82	+0.62	+0.62
Optuna per-bar XGBoost, TUNE_PER 25, best of 50 [baseline]	0.68	0.20	+0.048	+0.014	54.8	270	−0.61	−0.60	−1.02	+0.47	+0.47
Optuna per-bar XGBoost, TUNE_PER 25, best of 50, QLIKE-selected [baseline]	0.65	0.17	+0.046	+0.012	54.8	280	−0.64	−0.63	−1.07	+0.44	+0.44
Optuna per-bar LightGBM, TUNE_PER 25, best of 50, QLIKE-selected [baseline]	0.55	0.07	+0.039	+0.005	55.8	256	−0.73	−0.73	−1.27	+0.35	+0.35
Optuna per-bar random forest, TUNE_PER 25, best of 50, QLIKE-selected [baseline]	0.52	0.04	+0.037	+0.003	55.0	263	−0.77	−0.76	−1.31	+0.31	+0.31
Optuna per-bar random forest, TUNE_PER 25, best of 50 [baseline]	0.45	−0.03	+0.032	−0.002	54.6	270	−0.83	−0.83	−1.45	+0.24	+0.25
per-bar tree (Optuna, TUNE_PER=250, best-of-50)											
Optuna per-bar LightGBM, TUNE_PER 250, best of 50 [lean_vol_vix]	1.77	1.29	+0.125	+0.092	56.4	253	+0.49	+0.50	+1.01	+1.57	+1.57
Optuna per-bar LightGBM, TUNE_PER 250, best of 50 [all_features]	1.55	1.08	+0.110	+0.076	56.4	257	+0.27	+0.28	+0.61	+1.35	+1.35

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Forecast	Sharpe	Sharpe$^{\times}$	mean	mean$^{\times}$	hit %	buys	ΔSharpe vs R	ΔSharpe$^{\times}$ vs R	t vs R	ΔSharpe vs short	ΔSharpe$^{\times}$ vs short
Optuna per-bar XGBoost, TUNE_PER 250, best of 50 [lean_vol_vix]	1.52	1.03	+0.107	+0.074	57.3	265	+0.24	+0.23	+0.40	+1.31	+1.31
Optuna per-bar random forest, TUNE_PER 250, best of 50 [lean_vol_vix]	1.29	0.80	+0.091	+0.058	57.4	252	+0.01	+0.00	+0.02	+1.09	+1.07
Optuna per-bar XGBoost, TUNE_PER 250, best of 50 [all_features]	1.24	0.76	+0.088	+0.054	55.5	252	−0.04	−0.04	−0.09	+1.04	+1.04
Optuna per-bar XGBoost, TUNE_PER 250, best of 50 [baseline]	1.22	0.74	+0.087	+0.053	56.0	276	−0.06	−0.06	−0.12	+1.02	+1.02
Optuna per-bar LightGBM, TUNE_PER 250, best of 50 [baseline]	1.08	0.61	+0.077	+0.043	54.2	284	−0.20	−0.19	−0.41	+0.88	+0.88
Optuna per-bar random forest, TUNE_PER 250, best of 50 [all_features]	0.94	0.47	+0.067	+0.033	55.3	270	−0.34	−0.33	−0.57	+0.74	+0.74
Optuna per-bar random forest, TUNE_PER 250, best of 50 [baseline]	0.64	0.17	+0.046	+0.012	55.0	273	−0.64	−0.63	−1.09	+0.44	+0.44
per-bar tree (Optuna, TUNE_PER=1, best-of-25)											
Optuna per-bar LightGBM, TUNE_PER 1, best of 25 [all_features]	1.26	0.77	+0.089	+0.055	57.6	250	−0.02	−0.03	−0.04	+1.05	+1.04
Optuna per-bar LightGBM, TUNE_PER 1, best of 25 [lean_vol_vix]	1.25	0.77	+0.089	+0.055	57.5	261	−0.03	−0.03	−0.05	+1.05	+1.04
Optuna per-bar XGBoost, TUNE_PER 1, best of 25 [all_features]	1.13	0.65	+0.080	+0.046	57.5	253	−0.15	−0.15	−0.27	+0.93	+0.92
Optuna per-bar random forest, TUNE_PER 1, best of 25 [lean_vol_vix]	1.06	0.57	+0.075	+0.041	56.8	253	−0.22	−0.23	−0.39	+0.85	+0.85
Optuna per-bar XGBoost, TUNE_PER 1, best of 25 [baseline]	0.86	0.38	+0.061	+0.027	55.3	270	−0.42	−0.42	−0.71	+0.66	+0.66
Optuna per-bar XGBoost, TUNE_PER 1, best of 25 [lean_vol_vix]	0.81	0.33	+0.058	+0.024	55.4	255	−0.47	−0.47	−0.76	+0.61	+0.60
Optuna per-bar random forest, TUNE_PER 1, best of 25 [all_features]	0.57	0.09	+0.040	+0.007	55.2	255	−0.71	−0.71	−1.04	+0.37	+0.37
Optuna per-bar random forest, TUNE_PER 1, best of 25 [baseline]	0.54	0.06	+0.038	+0.004	53.8	273	−0.74	−0.74	−1.51	+0.33	+0.34
Optuna per-bar LightGBM, TUNE_PER 1, best of 25 [baseline]	0.46	−0.02	+0.033	−0.001	53.6	275	−0.82	−0.82	−1.42	+0.26	+0.26
per-bar tree (Optuna, TUNE_PER=25, best-of-25)											
Optuna per-bar random forest, TUNE_PER 25, best of 25 [lean_vol_vix]	1.31	0.82	+0.093	+0.059	56.7	266	+0.03	+0.02	+0.05	+1.10	+1.10
Optuna per-bar XGBoost, TUNE_PER 25, best of 25 [all_features]	1.22	0.74	+0.087	+0.053	56.4	259	−0.06	−0.06	−0.09	+1.02	+1.01
Optuna per-bar LightGBM, TUNE_PER 25, best of 25 [lean_vol_vix]	1.10	0.61	+0.078	+0.044	56.8	263	−0.18	−0.19	−0.33	+0.89	+0.89
Optuna per-bar XGBoost, TUNE_PER 25, best of 25 [lean_vol_vix]	1.09	0.61	+0.077	+0.044	56.6	259	−0.19	−0.19	−0.31	+0.89	+0.88
Optuna per-bar LightGBM, TUNE_PER 25, best of 25 [all_features]	1.08	0.59	+0.076	+0.043	56.8	243	−0.20	−0.21	−0.32	+0.88	+0.87
Optuna per-bar random forest, TUNE_PER 25, best of 25 [all_features]	0.80	0.32	+0.057	+0.023	55.8	260	−0.48	−0.48	−0.74	+0.60	+0.59
Optuna per-bar XGBoost, TUNE_PER 25, best of 25 [baseline]	0.73	0.25	+0.052	+0.018	55.3	264	−0.55	−0.55	−0.94	+0.53	+0.52
Optuna per-bar LightGBM, TUNE_PER 25, best of 25 [baseline]	0.53	0.05	+0.037	+0.004	54.8	268	−0.76	−0.75	−1.37	+0.32	+0.32
Optuna per-bar random forest, TUNE_PER 25, best of 25 [baseline]	0.33	−0.15	+0.023	−0.010	54.0	275	−0.95	−0.94	−1.69	+0.13	+0.13
per-bar tree (Optuna, TUNE_PER=1, best-of-10)											
Optuna per-bar LightGBM, TUNE_PER 1, best of 10 [lean_vol_vix]	1.57	1.08	+0.111	+0.077	57.2	254	+0.29	+0.29	+0.60	+1.36	+1.36
Optuna per-bar XGBoost, TUNE_PER 1, best of 10 [all_features]	1.51	1.02	+0.107	+0.073	57.5	265	+0.22	+0.22	+0.37	+1.30	+1.29
Optuna per-bar XGBoost, TUNE_PER 1, best of 10 [lean_vol_vix]	1.46	0.97	+0.103	+0.070	56.8	249	+0.18	+0.17	+0.30	+1.26	+1.24
Optuna per-bar random forest, TUNE_PER 1, best of 10 [all_features]	1.40	0.92	+0.099	+0.065	56.1	261	+0.12	+0.12	+0.20	+1.19	+1.19
Optuna per-bar LightGBM, TUNE_PER 1, best of 10 [all_features]	1.29	0.80	+0.091	+0.058	56.8	277	+0.01	+0.00	+0.01	+1.09	+1.08

(continued)

Forecast	Sharpe	Sharpe$^{\times}$	mean	mean$^{\times}$	hit %	buys	ΔSharpe vs R	ΔSharpe$^{\times}$ vs R	t vs R	ΔSharpe vs short	ΔSharpe$^{\times}$ vs short
Optuna per-bar random forest, TUNE_PER 1, best of 10 [lean_vol_vix]	1.10	0.61	+0.078	+0.044	56.4	263	−0.18	−0.19	−0.33	+0.89	+0.89
Optuna per-bar random forest, TUNE_PER 1, best of 10 [baseline]	0.79	0.31	+0.056	+0.022	54.5	275	−0.49	−0.48	−1.03	+0.59	+0.59
Optuna per-bar LightGBM, TUNE_PER 1, best of 10 [baseline]	0.73	0.25	+0.052	+0.018	54.4	278	−0.55	−0.54	−0.91	+0.53	+0.53
Optuna per-bar XGBoost, TUNE_PER 1, best of 10 [baseline]	0.59	0.11	+0.042	+0.008	54.4	274	−0.69	−0.69	−1.17	+0.39	+0.39
per-bar tree (Optuna, TUNE_PER=25, best-of-10)											
Optuna per-bar XGBoost, TUNE_PER 25, best of 10 [lean_vol_vix]	1.71	1.23	+0.121	+0.087	56.9	258	+0.43	+0.43	+1.00	+1.51	+1.51
Optuna per-bar LightGBM, TUNE_PER 25, best of 10 [lean_vol_vix]	1.67	1.18	+0.118	+0.084	56.9	268	+0.38	+0.38	+0.79	+1.46	+1.46
Optuna per-bar random forest, TUNE_PER 25, best of 10 [all_features]	1.58	1.11	+0.112	+0.078	56.8	263	+0.30	+0.31	+0.61	+1.38	+1.38
Optuna per-bar LightGBM, TUNE_PER 25, best of 10 [all_features]	1.50	1.02	+0.106	+0.073	56.6	273	+0.22	+0.22	+0.42	+1.30	+1.30
Optuna per-bar XGBoost, TUNE_PER 25, best of 10 [all_features]	1.48	0.99	+0.105	+0.071	55.9	269	+0.20	+0.20	+0.34	+1.27	+1.27
Optuna per-bar random forest, TUNE_PER 25, best of 10 [lean_vol_vix]	1.41	0.93	+0.100	+0.066	57.2	256	+0.13	+0.13	+0.24	+1.21	+1.20
Optuna per-bar LightGBM, TUNE_PER 25, best of 10 [baseline]	0.70	0.22	+0.050	+0.016	55.3	282	−0.58	−0.58	−1.00	+0.50	+0.50
Optuna per-bar XGBoost, TUNE_PER 25, best of 10 [baseline]	0.48	0.01	+0.034	+0.000	54.3	279	−0.80	−0.79	−1.35	+0.28	+0.28
Optuna per-bar random forest, TUNE_PER 25, best of 10 [baseline]	0.29	−0.19	+0.020	−0.014	54.0	267	−1.00	−0.99	−1.63	+0.08	+0.09
implied-vol representations											
per-bar elastic net [ivrep_term]	1.90	1.42	+0.134	+0.100	57.5	259	+0.62	+0.62	+1.35	+1.70	+1.69
per-bar ridge [ivrep_term]	1.87	1.38	+0.132	+0.098	58.1	262	+0.58	+0.58	+1.23	+1.66	+1.65
per-bar ridge [ivrep_vrp]	1.83	1.35	+0.129	+0.095	57.5	269	+0.55	+0.55	+1.01	+1.62	+1.62
per-bar lasso [ivrep_vrp]	1.66	1.18	+0.117	+0.084	56.5	268	+0.38	+0.38	+0.82	+1.45	+1.46
per-bar lasso [ivrep_term]	1.64	1.16	+0.116	+0.082	57.3	247	+0.36	+0.36	+0.77	+1.44	+1.43
per-bar ridge [ivrep_all]	1.61	1.13	+0.114	+0.080	56.7	264	+0.33	+0.33	+0.62	+1.41	+1.40
per-bar elastic net [ivrep_vrp]	1.59	1.11	+0.113	+0.079	56.8	259	+0.31	+0.31	+0.70	+1.39	+1.39
per-bar ridge [ivrep_target_scale]	1.58	1.10	+0.112	+0.078	56.5	256	+0.30	+0.30	+0.56	+1.38	+1.38
per-bar lasso [ivrep_innovations]	1.54	1.06	+0.109	+0.075	56.5	252	+0.26	+0.26	+0.52	+1.33	+1.33
per-bar lasso [ivrep_target_scale]	1.53	1.05	+0.108	+0.075	55.9	253	+0.25	+0.25	+0.50	+1.33	+1.33
per-bar ridge [ivrep_innovations]	1.50	1.02	+0.106	+0.072	56.8	267	+0.22	+0.22	+0.44	+1.30	+1.29
per-bar lasso [ivrep_all]	1.49	1.02	+0.106	+0.072	56.0	260	+0.21	+0.22	+0.45	+1.29	+1.29
per-bar elastic net [ivrep_all]	1.43	0.95	+0.101	+0.067	56.2	250	+0.15	+0.15	+0.29	+1.23	+1.22
per-bar elastic net [ivrep_innovations]	1.40	0.91	+0.099	+0.065	56.9	242	+0.12	+0.11	+0.22	+1.19	+1.19
per-bar elastic net [ivrep_target_scale]	1.25	0.77	+0.088	+0.055	55.9	249	−0.03	−0.03	−0.07	+1.05	+1.04
Table B: scored on the forecast's own days, paired with the reference on them (550 trade days from 2022-01-03)											
chain-period buckets											
per-bar elastic net [ivrep_slice_slope], 500 sessions	2.13	1.75	+0.142	+0.118	59.5	148	+0.52	+0.51	+0.92	+1.49	+1.48
per-bar ridge [ivrep_slice_slope], 500 sessions	2.09	1.71	+0.139	+0.115	58.9	165	+0.48	+0.48	+0.89	+1.45	+1.45
per-bar lasso [ivrep_slice_slope], 500 sessions	1.90	1.53	+0.127	+0.103	59.8	154	+0.30	+0.29	+0.52	+1.26	+1.26
per-bar ridge [ivslice_only], 500 sessions	1.35	0.98	+0.091	+0.067	58.5	143	−0.25	−0.26	−0.43	+0.71	+0.71
per-bar elastic net [ivslice_only], 500 sessions	1.33	0.96	+0.089	+0.065	58.7	140	−0.28	−0.28	−0.43	+0.69	+0.69
per-bar lasso [ivslice_only], 500 sessions	1.08	0.71	+0.072	+0.048	57.8	141	−0.53	−0.53	−0.70	+0.44	+0.44
per-bar lasso [mfiv_only], 500 sessions	1.00	0.63	+0.067	+0.043	57.6	148	−0.61	−0.61	−0.79	+0.35	+0.36
per-bar elastic net [mfiv_only], 500 sessions	0.97	0.60	+0.065	+0.041	57.1	147	−0.64	−0.63	−0.85	+0.33	+0.34
per-bar ridge [mfiv_only], 500 sessions	0.71	0.35	+0.048	+0.024	57.1	155	−0.90	−0.89	−1.33	+0.07	+0.08

Table 3: VRP sizing on the deck's scorer, 865 days from the second deck day, midpoint fill. The size on day t uses every earlier deck day back to the first: robust clip $q_t = \mathrm{clip}(s_t/\tilde{m}_t, -1, 1)$ with $\tilde{m}_t$ the median of $|s|$ on the earlier days, and the rank of $|s_t|$, sign kept. Δ is the rule's Sharpe minus sign(s) on the same days (point estimate). These Sharpes are the deck's own scorer, not the plain-forecast Sharpes of the trade table.

forecast	sign(s)	robust clip		rank	
	Sharpe	Sharpe	Δ vs sign(s)	Sharpe	Δ vs sign(s)
per-bar ridge [`baseline`]	1.38	1.19	−0.19	1.17	−0.21
per-bar ridge [`all_features`]	1.73	1.57	−0.15	1.39	−0.33
per-bar lasso [`all_features`]	2.02	1.67	−0.35	1.53	−0.49
per-bar elastic net [`all_features`]	2.06	1.61	−0.45	1.61	−0.45

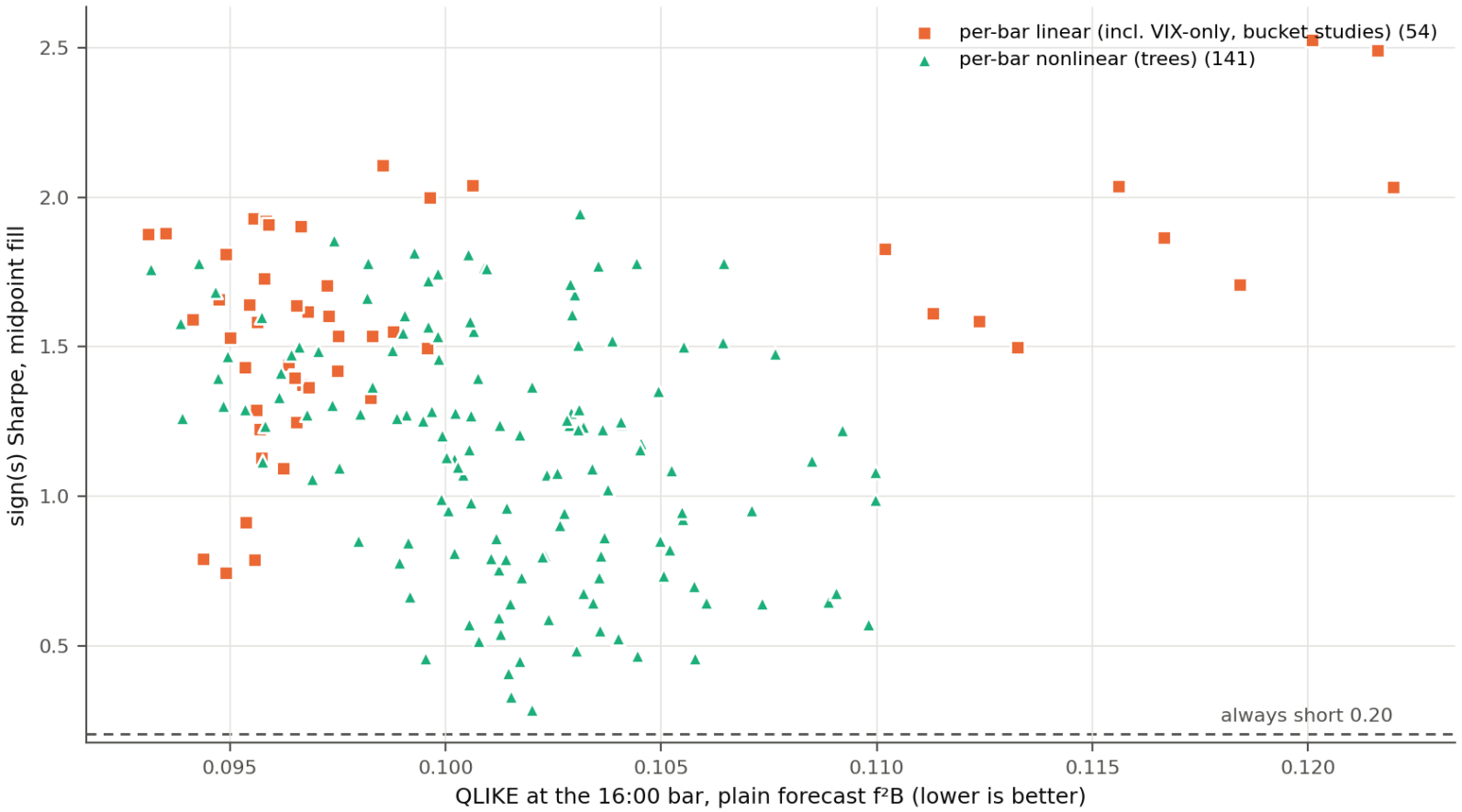


Figure 1: 16:00 QLIKE of the plain forecast f^2B against the sign(s) Sharpe ratio (midpoint fill), table A (check rows and identical forecasts left out); the dashed line is always short. R = reference, H = recommended headline. 1 forecasts lie beyond the plotted QLIKE range (upper quartile + 3 interquartile ranges = 0.123) and are not drawn: per-bar ridge [free_feasible] (QLIKE 0.143, Sharpe 2.08).